

Dingxin Tao

AI Native Software Engineer

Full Stack · AI Applications & Agents · Solution Delivery

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Proficient with Codex, Claude and Cursor for requirements, context preparation, prototyping, implementation, code review and regression checks. Builds software for personal learning, AI video creation, local AI assistance and business transactions. Internship experience includes internal ERP and AI image services; communicates in English and Mandarin.

EDUCATION

University of Southern California · Viterbi School of Engineering

Master of Science in Analytics

Jan 2026 - May 2027 (expected)

University of California, Irvine

Bachelor of Science in Computer Science

Sep 2023 - Sep 2025

EXPERIENCE

China Quality Certification Centre

Jun 2026 - Jul 2026

New Energy Department · ERP & Data Analytics Development Intern

- Developed the New Energy Department's internal ERP and data analytics system.

Synergies

Sep 2025 - Dec 2025

Software Engineering Intern

- Built image processing services with the OpenAI Vision API; fine-tuned Stable Diffusion and ran A/B experiments to compare generation configurations.
- Developed Flask REST APIs with JWT authentication, rate limiting and Swagger documentation; deployed services with Docker and stored images in AWS S3.
- Worked with product and design teams to refine user flows through behavior analysis and reusable React components.

PROJECTS

Conuo · Learning Workspace

2026 · Ongoing

Next.js / React / TypeScript · PDF.js · Tiptap · IndexedDB

- Built a responsive web workspace connecting PDF, DOCX, Markdown and TXT imports, document reading, text highlighting and Tiptap note editing.
- Implemented content fingerprints, source versions and scoped keyword retrieval with page / section citations; persisted workspaces and originals in IndexedDB, with revision recovery and ZIP export.
- Prepared Supabase Auth, private Storage interfaces and database migrations to establish account and data access boundaries for cloud integration.

MultiMix · AI Video Workspace

2026 · Ongoing

Next.js / TypeScript · FastAPI / Python · PostgreSQL · Redis / RQ

- Connected creative conversations, storyboards, asset retrieval, narration and video export; integrated and extended a browser editor so generated results remain editable and persistable.
- Implemented asynchronous generation jobs, status updates and recovery, with semantic asset search, source validation and export checks for long-running tasks and external services.
- Diagnosed editor persistence, generation failures and confirmation flows; added human confirmation, prompt-injection safeguards and model output validation.

PROJECTS (CONTINUED)

OPC Workspace · Local AI Desktop App

2026 · Ongoing

React / TypeScript · Tauri · Go · SQLite

- Built workflows for tasks, projects, customers and deliverable review, with migrations, backup / restore, local persistence and Windows test installers.
- Integrated OpenAI Chat Completions and Anthropic Messages protocols into an AI runtime with streaming responses, session memory and context compaction.
- Connected workspace data through restricted tools with write previews, human confirmation and execution traces; added text / PDF ingestion, citation validation and local evaluations.

As-a · Multilingual Services Marketplace

2026 · Ongoing

Next.js / TypeScript · Supabase Auth / PostgreSQL / Storage / Realtime

- Built Arabic / English interfaces with RTL support for provider onboarding, review, conversations, orders and delivery; isolated business data through row-level security and roles.
- Implemented server-side payment verification, duplicate-callback protection, idempotent settlement, admin audit logs and withdrawal reservations; added AI support APIs with authentication, shared rate limits, validation and timeouts.

Vibio SEO · Reusable AI Workflows

2026 · Ongoing

Python runtime · JSON Schema · Skill packaging · Offline regression

- Built executable skills for SEO planning, audits, fixes, content and keyword work, using Python tools to analyze search data and produce structured reports.
- Defined JSON Schema output contracts, self-contained skill packages, reference integrity and drift checks, and offline regressions for repeatable workflow execution and validation.

LEADERSHIP & COLLABORATION

UCI CareTech

Jun 2024 - Sep 2025

Marketing Lead · Campus Organization

- Led outreach strategy, website content and SEO for an AI health technology project; supported its responsive React / TypeScript website and coordinated technical and marketing work.

TECHNICAL & DELIVERY SKILLS

AI Native development Codex, Claude, Cursor; AI Agent, RAG, Workflow.

Applications & engineering Python, TypeScript / JavaScript, SQL; React, Next.js, FastAPI, Flask; REST APIs, Git, Docker.

Data & infrastructure PostgreSQL, SQLite, Redis, Supabase, IndexedDB, AWS S3 / EC2; authentication, data access, asynchronous jobs, backup / restore.

AI & analytics Model APIs, tool use, streaming, citation validation; Pandas, NumPy, Scikit-learn, PyTorch, Stable Diffusion.

Collaboration & delivery English and Mandarin; business requirements, prototyping, API integration, debugging and validation.